Year 13 Maths - Further integration

Similar questions - Integration by substitution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Use the substitution $u^2 = x^2 + 1$ to show that $\displaystyle\int_0^1 \frac{x^3}{\sqrt{x^2+1}}\,\mathrm{d}x = \frac{2-\sqrt{2}}{3}$. **[6]**

(b) Use the substitution $x = 2\sin\theta$ to find the exact value of $\displaystyle\int_0^{\sqrt{2}} \sqrt{4-x^2}\,\mathrm{d}x$. **[6]**

2

(a) Use the substitution $t = \sqrt{x+1}$ to show that $\displaystyle\int_0^3 \mathrm{e}^{\sqrt{x+1}}\,\mathrm{d}x = \int_1^2 2t\,\mathrm{e}^t\,\mathrm{d}t$. **[3]**

(b) Hence show that $\displaystyle\int_0^3 \mathrm{e}^{\sqrt{x+1}}\,\mathrm{d}x = 2\mathrm{e}^2$. **[4]**

Question 1

(a) $2u\,\mathrm{d}u = 2x\,\mathrm{d}x$, so $x\,\mathrm{d}x = u\,\mathrm{d}u$ and $x^2 = u^2 - 1$. Limits $u = 1$ to $\sqrt{2}$. The integral becomes $\displaystyle\int_1^{\sqrt{2}} \frac{(u^2-1)u}{u}\,\mathrm{d}u =$ $\left[\frac{u^3}{3} - u\right]_1^{\sqrt{2}} = \left(\frac{2\sqrt{2}}{3} - \sqrt{2}\right) - \left(\frac{1}{3} - 1\right) = \frac{2-\sqrt{2}}{3}$

(b) $\mathrm{d}x = 2\cos\theta\,\mathrm{d}\theta$, $\sqrt{4 - 4\sin^2\theta} = 2\cos\theta$. Limits 0 to $\frac{\pi}{4}$. $\displaystyle\int_0^{\pi/4} 4\cos^2\theta\,\mathrm{d}\theta = \int_0^{\pi/4} (2 + 2\cos 2\theta)\,\mathrm{d}\theta = [2\theta + \sin 2\theta]_0^{\pi/4} =$ $\frac{\pi}{2} + 1$

Question 2

(a) $x = t^2 - 1$, $\mathrm{d}x = 2t\,\mathrm{d}t$; $x = 0 \Rightarrow t = 1$, $x = 3 \Rightarrow t = 2$, giving $\displaystyle\int_1^2 2t\,\mathrm{e}^t\,\mathrm{d}t$

(b) By parts with $u = 2t$, $v' = \mathrm{e}^t$: $[2t\mathrm{e}^t]_1^2 - \displaystyle\int_1^2 2\mathrm{e}^t\,\mathrm{d}t = 4\mathrm{e}^2 - 2\mathrm{e} - 2\mathrm{e}^2 + 2\mathrm{e} = 2\mathrm{e}^2$